MIME: Multi-Strategy Imbalance Mitigation Engine for Skin Cancer Classification

Md. Nasif Safwan[1][0009−0002−8098−4355], Souhardo Rahman[1][0009−0007−3774−0116], Md. Abdulla Al Hasib[1][0009−0002−6326−0419], Md. Hasibul Islam[1][0009−0004−7864−1657], Mahamodul Hasan Mahadi[1][0009−0002−1601−8793], and Mohammad Nasir Uddin[2][0000−0001−5331−3906]

[1] Department of Computer Science, American International University-Bangladesh, Dhaka, Bangladesh
[2] Department of Electrical and Electronic Engineering, American International University-Bangladesh, Dhaka, Bangladesh

Abstract. Severe class imbalance is an major obstacle in automated skin lesion diagnosis, where minority classes such as squamous cell carcinoma are often underdetected. We propose MIME, a unified framework that evaluates three complementary strategies: sampling to rebalance the training pool, loss functions to adjust learning priorities, and DC-GAN augmentation to enrich minority cases. Built on an EfficientNetV2-S backbone with custom classifier heads and optional channel attention, the MIME framework provides a controlled setting to assess each method individually and in combination. Experiments on the ISIC 2019 dataset demonstrate that sampling and loss based adjustments yield moderate benefits, but generative augmentation delivers the most substantial improvements. Compared to a linear baseline, accuracy increased from 82.25% to 90.08%, macro F1 increased from 60.42% to 82.84%, and SCC F1 improved dramatically from 11.89% to 53.18%. These results highlight the importance of targeted strategies for minority class enhancement, showing that DC-GAN based augmentation can significantly boost sensitivity to aggressive but underrepresented lesions while maintaining strong performance on majority classes. Through a unified and reproducible protocol, MIME provides a practical foundation for developing transparent, more robust, and clinically reliable dermatology AI systems.

Keywords: Skin cancer · Data imbalance · Generative augmentation · Image processing · Deep learning.

1 Introduction

Skin cancer is the most commonly occurring malignancy worldwide, with reported cases exceeding 1.5 million in 2022 [1]. The majority arise from Non Melanoma Skin Cancers (NMSCs), mainly Basal Cell Carcinoma (BCC) and Squamous Cell Carcinoma (SCC), which remain a principal cause of mortality

associated with skin cancer [22]. Among them, cutaneous SCC is the second most common after BCC and is responsible for tens of thousands of deaths annually [4]. Early detection dramatically improves outcomes. For example, the five year survival rate for melanoma exceeds 99% when caught early but drops to around 35% after distant metastasis [22]. This sharp decline underscores the critical need for timely and reliable diagnostic tools. In recent years, advances in Artificial Intelligence (AI), particularly in Deep Learning (DL), have transformed dermatological diagnostics. Convolutional Neural Network (CNN) now achieve performance on par with dermatologists in classifying both dermoscopic and clinical images [17], paving the way for AI-assisted systems that can augment clinical workflows and extend care access. In teledermatology, such tools are especially valuable for rural or underserved communities with limited access to specialists [16].

However, a persistent barrier to clinical reliability is the severe class imbalance present in skin lesion datasets. Whether in benchmark datasets or real world settings, benign lesions vastly outnumber malignant cases, particularly SCC, leading models to bias toward majority classes [8]. As a result, sensitivity to rare but critical conditions is often compromised, undermining the trustworthiness of AI systems in high stakes applications [18]. To address this challenge, we introduce the Multi strategy Imbalance Mitigation and Evaluation (MIME) framework, a unified approach designed to systematically tackle class imbalance in skin cancer classification through a blend of complementary strategies. The main contributions of this work are as follows:

- We present MIME, a unified framework integrating data level sampling, algorithm level weighting, and generative modeling with Deep Convolutional GAN (DC-GAN).
- MIME enables controlled comparisons of individual and combined effects with a reproducible benchmarking protocol.
- We apply MIME to the classification of skin lesion images with a particular focus on clinically significant minority classes such as SCC, demonstrating improved sensitivity and overall robustness.

By integrating diverse mitigation strategies under a single evaluation pipeline, MIME sets the stage for developing fairer, more reliable diagnostic tools in dermatology.

2 Related Work

Recent literature on automated skin cancer classification shows a marked shift toward hybrid and ensemble approaches, while continuing to grapple with the persistent challenge of data imbalance in Melanoma (MEL), BCC and SCC classes. Researchers have moved beyond single purpose architectures. Abohashish, Amin and Elsedimy explored a hybrid LSTM CNN to capture spatial dependencies, while Mustafa et al. coupled ResUNet++ segmentation with an AlexNet Random Forest classifier to enhance robustness [2,10]. Ozdemir

and Pacal further advanced this trend by combining ConvNeXtV2 with separable self attention to bridge the strengths of CNN and Vision Transformers [13]. Ensemble designs are also gaining ground, as in Halder et al. with their fuzzy rank based framework using multiple pre trained models, or Mehmood et al. with SBXception, a shallower and broader variant that challenged the dominance of very deep networks [6, 9].

Transfer learning has also become central to recent progress, with researchers adapting a variety of pre trained networks to dermatological imaging. Venugopal et al. refined EfficientNet variants for malignant–benign classification, while Yilmaz et al. explored lighter architectures including MobileNet and NASNetMobile through careful fine tuning [20, 21]. Islam and Panta broadened the comparison further by evaluating ResNet, DenseNet and Inception families, underscoring how transfer learning enables robust performance even when medical datasets are limited [7]. A unifying concern across studies is the severe imbalance in benchmark datasets such as ISIC 2019 [5]. MEL and BCC dominate, while SCC and other minority classes remain underrepresented, introducing bias during training and masking deficiencies in clinical performance.

To counter this, researchers have relied on a spectrum of strategies. Traditional augmentation methods, including flips and rotations, remain common and have been applied by Nawaz et al., Musthafa et al. and Arshad et al. Generative approaches are also gaining traction, with Saeed et al. showing how GAN based augmentation can strengthen detection of minority classes [3, 11, 12, 15]. Beyond data centric solutions, algorithmic techniques called SMOTE Tomek sampling have been explored by Tahir et al. [19]. More recent work has sought to combine these perspectives, as in Arshad et al. [3] hybrid contrast enhancement with class balancing augmentation, or in Gilani et al. [14] experiments with spiking networks, both of which suggest new directions for handling imbalance. Important limitations remain. Strong benchmark results often fail to hold up in real world use, with many models struggling to generalize across populations and imaging conditions. Tahir et al. highlighted demographic bias, noting that their system worked well only on fair toned skin [19]. Interpretability is another challenge. As Ozdemir and Pacal observed, attention based models can achieve high accuracy yet still act as black boxes, limiting clinician trust [13].

Building on these findings, We introduce the MIME framework, which combines data-level, algorithm-level, and generative techniques in a unified framework. MIME enables controlled comparison of these strategies and offers a reproducible path toward fairer and more reliable skin cancer classification.

3 Methodology

This study introduces a structured evaluation framework to systematically assess the effectiveness of class imbalance mitigation techniques in dermatological image classification. Instead of relying on isolated methods we developed the

MIME framework, a controlled pipeline that isolates and quantifies the impact of each strategy both individually and in combination.

3.1 Dataset Acquisition & Data Transformation

This study uses the ISIC 2019 dataset [5] and focuses only on three malignant skin cancer types, namely MEL, BCC, and SCC, to emphasize cancer detection over general lesion classification.The training set consists of 6,778 dermoscopic images, including 3,624 MEL, 2,662 BCC, and only 492 SCC cases, reflecting a significant class imbalance. A stratified validation set of 4,298 images maintains this distribution, and a separate test set of 793 images is used for final evaluation.

We use a layered augmentation pipeline that keeps images realistic while adding controlled variation. Each image is resized to 256×256, cropped to 224×224, converted to tensors, and normalized with ImageNet statistics. To improve robustness, we apply horizontal flips ($p = 0.5$), vertical flips ($p = 0.3$), rotations up to $\pm 20^{\circ}$, light color jitter, and random erasing ($p = 0.1$). These augmentations mimic variations in dermoscopy including tilt, lighting change, and hair occlusion, guiding the model to focus on lesion shape and texture.

3.2 Neural Design of MIME

Base Network Architecture We used EfficientNetV2-S as the backbone for its efficiency and detailed feature extraction. ImageNet pretraining offered a strong base, and the classification head was replaced with custom modules to examine how model capacity affects imbalance handling.

Classifier Head Configurations To analyze how architectural complexity influences imbalance handling, three classifier head designs were evaluated, each offering a different balance of expressiveness, regularization, and cost.

Linear Projection: A single linear layer mapping the 1280 dimensional feature vector to three output classes. With no activation or regularization, this served as a minimal baseline to isolate imbalance handling effects.

Single Stage Filter: Included a hidden layer with 256 neurons, ReLU activation, dropout at 0.3, and an output layer. This moderate design allowed non linear learning while controlling overfitting.

Multi Layer Projection: Used three fully connected layers ($1280 \rightarrow 512 \rightarrow 256 \rightarrow 3$), batch normalization after each intermediate layer, and staged dropout rates (0.3, 0.24, 0.18). This deeper architecture enhanced pattern learning, especially for underrepresented lesion types.

Attention Driven Feature Enhancement To enhance the model's focus on diagnostically relevant cues, a channel wise attention module was incorporated into selected configurations. Given a backbone feature vector $\mathbf{f}_{\text{backbone}} \in \mathbb{R}^{1280}$, the attention weights were computed as shown in Equation 1.

$$\boldsymbol{\alpha} = \sigma\left(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1 \cdot \mathbf{f}_{\text{backbone}} + \mathbf{b}_1) + \mathbf{b}_2\right), \quad \mathbf{f}_{\text{attended}} = \boldsymbol{\alpha} \odot \mathbf{f}_{\text{backbone}} \tag{1}$$

$$\alpha_i = \sigma\left(\sum_{j=1}^{320} W_{2_{ij}} \cdot \mathrm{ReLU}\left(\sum_{k=1}^{1280} W_{1_{jk}} f_k + b_{1_j}\right) + b_{2_i}\right) \tag{2}$$

where $\mathbf{W}_1 \in \mathbb{R}^{320\times1280}$, $\mathbf{W}_2 \in \mathbb{R}^{1280\times320}$, and σ denotes the sigmoid activation. The first layer reduced dimensionality ($1280 \rightarrow 320$), followed by ReLU, while the second layer restored it and applied a sigmoid gate to produce normalized weights. This compression–expansion design modeled channel dependencies with low overhead. Each coefficient was computed as shown in Equation 2.

By selectively amplifying or suppressing feature channels, the attention module improved sensitivity to subtle, class discriminative patterns, particularly benefiting underrepresented lesion types.

4 Implementation Details

Class imbalance is a major challenge in dermatological image analysis. In the dataset MEL dominates while SCC is rare, causing models to favor majority classes and miss critical lesions. To address this, we explored three strategies: adjusting the training distribution, refining the loss function, and generating synthetic samples, offering insight into their individual and combined impact on minority class detection. The overall design is illustrated in Figure 1.

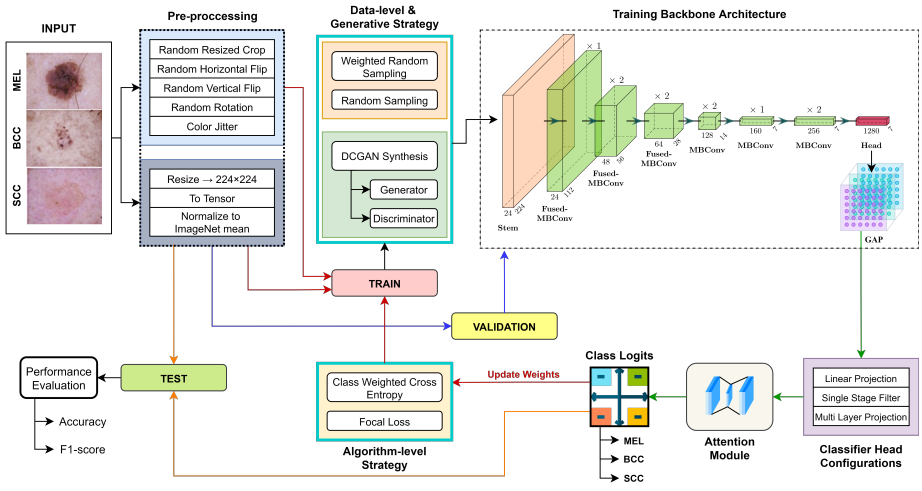


Fig. 1: Proposed MIME Framework for Skin Cancer Classification

4.1 Sampling Strategies

We explored two data level techniques to mitigate class imbalance. In weighted random sampling, each training sample was assigned a selection probability based on the inverse frequency of its class. This gave underrepresented classes greater exposure during training. In the dataset the computed class weights were 0.74 for MEL, 1.0 for BCC, and 5.3 for SCC, meaning SCC samples were selected roughly seven times more often than MEL.

Random oversampling balanced the dataset by duplicating minority class samples to match the size of the majority class. SCC and BCC samples were replicated approximately 7.2 and 1.4 times, respectively, while MEL required no duplication. This increased the training set from 8,473 to around 13,566 images, ensuring equal representation of all three classes across training epochs.

4.2 Loss Functions

To tackle class imbalance, we tested two loss functions independently: weighted cross entropy and focal loss. Weighted cross entropy assigned higher penalties to BCC and SCC, giving them greater influence during training. On the other hand, focal loss, with $\gamma = 2.0$ and class weights α, emphasized hard to classify minority cases by reducing the impact of easy examples. This encouraged sharper decision boundaries and improved recognition of rare lesions.

4.3 Generative Modeling: DC-GAN Based Synthesis

To overcome the shortage of SCC samples, we used a DC-GAN to create additional realistic images only for this class. The choice was based on the fact that SCC had just 492 images, compared to 3,624 in MEL and 2,662 in BCC. These generated images expanded the minority class without reducing diagnostic validity. The generator transformed 128 dimensional Gaussian noise vectors ($z \sim \mathcal{N}(0, I)$) into synthetic RGB images of size 224×224×3, while the discriminator learned to distinguish real sample from generated samples. Training followed the standard adversarial objective shown in Equation 3.

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))] \quad (3)$$

The generator used transpose convolutions with batch normalization and ReLU activations, ending with a tanh layer to match image intensity ranges. The discriminator mirrored this architecture with downsampling blocks, Leaky ReLU (slope $\alpha = 0.2$), and batch normalization, outputting a scalar real or fake prediction. We trained both networks for 120 epochs using Adam (learning rate=0.0002, β_1=0.5, β_2=0.999), batch size 16, and a 1:1 generator-discriminator update ratio. Total of 2000 synthetic images were generated for SCC class for data imbalance mitigation. As the synthetic images were included in model training only, it was important to ensure their reliability and diagnostic relevance. To ensure reliability, FID and SSIM techniques were applied to verify

the closeness to real samples and the retention of skin cancer lesion structure. Figure 2 presents representative examples of original and GAN-generated SCC images, qualitatively validating the realism and diversity of the synthesized samples. By adding synthetic yet realistic samples to the SCC class, the approach mitigated class imbalance and may lead to better classification performance.

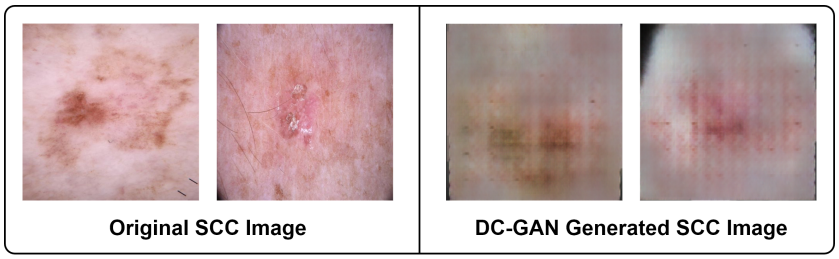


Fig. 2: Original vs DC-GAN generated SCC images

5 Results and Discussion

Table 1 summarizes the results on a stratified test set of 793 dermoscopic images. Among all methods, the DC-GAN based approach delivered the highest performance, achieving 90.08% test accuracy and 82.84% macro F1 score, setting a new benchmark for imbalanced skin lesion classification.

Table 1: Performance metrics for different methods

Method	Accuracy (%)	F1 score (%)				
		Weighted	Macro	MEL	BCC	SCC
Linear Projection [Baseline]	82.25	80.29	60.42	88.02	81.35	11.89
Single Stage Filter	87.84	87.55	77.16	92.30	87.08	52.11
Multi Layer Projection	87.52	87.46	77.03	92.39	86.77	51.92
Weighted Sampling	87.72	87.33	76.44	92.33	86.81	50.18
Random Oversampling	87.60	87.12	76.52	92.06	86.52	50.97
Class Weighted Loss	86.66	86.41	75.14	91.50	86.01	47.92
Weighted Cross Entropy Loss	87.15	86.82	75.70	92.06	86.10	48.94
DC-GAN	**90.08**	**89.93**	**82.84**	**92.13**	**87.44**	**53.18**

Table 2: Relative improvements and training time over baseline

Method	Accuracy	Macro F1	SCC F1	SCC-Improve	Time(s)
Single Stage Filter	↑6.78%	↑27.6%	↑338.1%	4.38×	7483.33
Multi Layer Projection	↑6.42%	↑27.4%	↑336.8%	4.37×	8921.45
Weighted Sampling	↑6.65%	↑26.5%	↑322.0%	4.22×	9060.27
Class Weighted Loss	↑5.36%	↑24.3%	↑302.9%	4.03×	9062.20
Weighted Cross Entropy Loss	↑5.95%	↑25.3%	↑311.6%	4.12×	8906.21
Random Oversampling	↑6.50%	↑26.6%	↑328.7%	4.29×	11487.23
DC-GAN	**↑9.63%**	**↑37.1%**	**↑347.2%**	**4.47×**	**14449.89**

Table 2 quantifies the relative improvements achieved by each approach compared to the baseline, demonstrating the substantial gains in minority class detection. To quantify the performance gains over a baseline model, we

use two complementary metrics. Relative improvement expresses the gain as a percentage, while relative increase represents it as a multiplicative factor.

The results underscore the clear advantage of generative augmentation in addressing class imbalance in dermatological AI. DC-GAN outperformed all other methods, achieving 90.08% accuracy and 82.84% macro F1 score. Most notably, it improved detection of the minority SCC class from a baseline F1 score of just 11.89% to 53.18%, a remarkable 347% relative gain. This improvement directly tackles one of the most pressing challenges in skin cancer classification of reliably identifying aggressive but underrepresented subtypes.

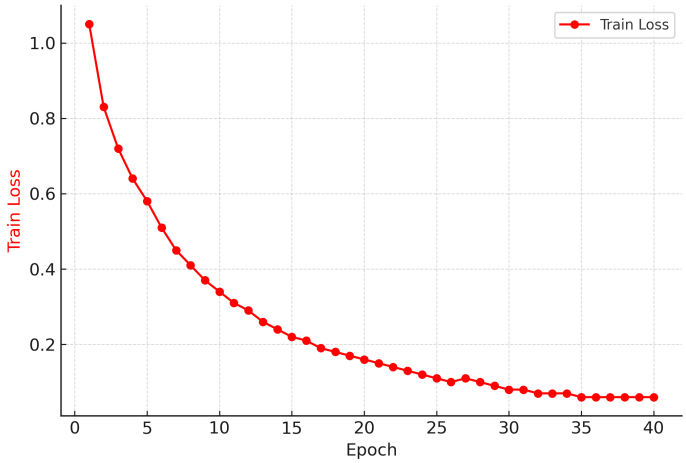


Fig. 3: Training loss curve showing a steady decrease in loss over 40 epochs

Figure 3 illustrates the training loss progression over 40 epochs. The loss dropped sharply during the initial epochs and continued to decline gradually afterward, eventually stabilizing below 0.1. This consistent downward trend indicates efficient optimization and strong convergence of the model. DC-GAN achieved the best performance-to-time ratio at 14.91×10^{-5} accuracy per second, over twice that of random oversampling. Even architectural refinements alone boosted macro F1 from 60.42% to 77.16%, underscoring the value of combining model design with imbalance mitigation.

Table 3: Performance comparison and improvement, Baseline vs. DC-GAN

Method	**Accuracy (%)**	**F1 score (%)**				
		Weighted	**Macro**	**Class-MEL**	**Class-BCC**	**Class-SCC**
Linear Projection	82.25	80.29	60.42	88.02	81.35	11.89
DC-GAN	**90.08**	**89.93**	**82.84**	**92.13**	**87.44**	**53.18**
Improvement	↑7.83	↑9.64	↑22.42	↑4.11	↑6.09	↑41.29

Table 3 shows that the DC-GAN strategy not only improved overall accuracy but also brought a major leap in detecting the underrepresented SCC class. This

marks a shift from struggling with rare lesions to achieving balanced performance across categories while maintaining strong reliability for MEL and BCC.

6 Conclusion

This study introduced MIME, a unified framework for addressing severe class imbalance in skin lesion classification. Through a structured evaluation of data level, algorithm level, and generative methods, we found that conventional approaches including class weighting and oversampling provide only marginal benefits. In contrast, the integration of synthetic data generated using DC-GAN led to a marked improvement in performance, increasing the F1 score for SCC from 11.89% to 53.18%, along with notable gains in macro F1 and overall accuracy. These findings show that targeted augmentation is effective in low-resource clinical settings, especially with refined architectures and attention mechanisms. Although MIME suits low-resource settings, training DC-GAN still requires considerable computation and results may reflect dataset specific biases. MIME's principles could be extend to other imbalanced medical datasets for example, diabetic retinopathy or rare tumor detection. In general, MIME defines imbalance mitigation as a foundational requirement rather than a secondary concern in the development of reliable medical AI systems.

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